Agroforestry systems as production-integrated compensation

An assessment proposal based on woody biotopes with official values

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IN BRIEF

- The biotope value list of Rhineland-Palatinate contains no biotope type for agroforestry. The proposal places agroforestry on the scale of 0 to 24 points by analogy with hedges and tree rows.
- The point levels follow value intervals of the official bases (own derivations: Section 8): tree strip core 8 to 18 points, along a watercourse up to 19; fringe (herbaceous strip along the trees) 16 points. All value-determining features are laid down in the planting plan and can be checked.

1 The problem

The biotope value list contains no biotope type for agroforestry, which under agricultural law counts as land management (§ 4(2) GAPDZV). The closest type, energy wood crop (HJ9), has the value of intensively used arable land at 6 points, while comparable woody biotopes score 8 to 19 points (MKUEM, 2021, Annex 7.1). So far, agroforestry can only be assessed case by case, by analogy and within the discretionary range of at most 3 points; the proposal makes this analogy checkable.

2 Analogy approach and anchors

The starting point is biotope types with a fixed base value, the anchors: tree group and tree row (BF1 to BF6) as well as shrub hedge, woody strip, and tree hedge (BD2, BD3, BD6), each predominantly non-autochthonous in young form, at 8 points. The woody strip (BD3), planted and managed at regular intervals, is the closest equivalent (LökPlan GbR, 2024, p. 40). Apart from the author's own derivations listed in Section 8, the point increments follow value intervals of the list: the age stages of small woody biotopes and the upgrade criteria of the forest biotope types (MKUEM, 2021, pp. 12–13).

3 Requirements for the basic form

- The initial site is arable land, grassland, or a permanent crop.
- The agroforestry definition and the requirements of eco-scheme 3 (ÖR 3, the CAP payment for maintaining agroforestry) are met (§ 4(2) and Annex 5, no. 3 GAPDZV): at least 2 tree strips, each at most 25 m wide, covering 2 to 40% of the area, 20 to 100 m apart. Eco-scheme 3 is not legally required for compensation measures; the proposal uses its design requirements as a quality standard, also on permanent crops, to which eco-scheme 3 itself does not apply.
- No fertilisation of the tree strips; plant protection products at most in the first year after planting and only in agreement with the lower nature conservation authority.
- Minimum retention period 20 years; 50 years for standard trees (Überhälter, trees left standing when the rest is harvested).
- Areas for resting birds and open-land breeding birds: as a rule not suitable, because trees can impair the conservation objectives; legally protected biotopes only after site-specific consultation with the lower nature conservation authority.

The tree strip core and the fringe are assessed. The cultivated alley farmed between the strips keeps its arable or grassland code. The biotope type group is called agroforestry woody vegetation (collective code BC0): BC1 linear, BC3 riparian, up to 19.

4 Value axes with thresholds

Value axis	Points	Threshold, to be laid down in the planting plan	Link to the anchor	Ecological effect
Basic form	**8**	Requirements as in Section 3	hedge, tree row; young, non-autochthonous	linear trees and shrubs in the agricultural landscape
Harvest mosaic	**+1**	at least 3 sections harvested at different times; at least one is left standing at each harvest	forest criterion verticality, age threshold adjusted	sections of different ages, multi-layered stand
Multi-row structure	**+1**	predominantly at least 4 rows or 10 m width per tree strip	approximation to the forest edge (17), smallest step	fewer inputs from the field edge, more niches (Kramer et al., 2026)
Species diversity	**+2**	at least 3 species native to the site or of particular ecological value, making up 10% of all trees as standard trees or 50% of the understorey	forest criteria tree species diversity and rare species, +1 each	native fauna depends mainly on native trees and shrubs (Kramer et al., 2026)
extended harvest interval	**+3**	harvest interval of 10 years or more on at least 20% of the tree strip area	age step young to medium	trees remain in place for longer
or standard trees	**+6**	at least 10% of all trees kept for 50 years, over 20% of canopy cover in the target state	age step young to old; canopy cover rule (BfN & BMU, 2021)	old cavity trees: key structures for wood-dwelling beetles (Horák, 2017)
Along a watercourse, BC3	**+1**	along at least 80% of the bank line, at most 5 m landward of the top of the bank	riparian woodland old (19) minus small woody biotope old (18)	retention of nutrients and sediment (Kumwimba et al., 2024)
Cap	**18 / 19**	Maximum value of the anchor: old autochthonous hedge 18, old riparian woodland 19. The full form reaches it exactly; the cap keeps the link to the anchor visible.		

Harvest interval (thesis: rotation period) and standard trees are mutually exclusive. The shortened age scale for fast-growing stands is taken from the mapping guide for the BKompV (Tschiche et al., 2025, p. 208). The thresholds apply per field.



Figure 1: Value axes on the anchor, capped at its maximum value; fringe assessed separately. Grey on the point scale: guide values of the initial state (arable land 6, species-rich grassland 14). Own illustration.

5 Fringe as a separate sub-area

The herbaceous, flowering, or tall-forb strip along the trees receives 16 points, like near-natural fringe strips and field and grassland margins (MKUEM, 2021, Annex 7.1). It is added, not averaged, because fringe and trees differ in vegetation, value, and development time. It establishes itself within a few years and therefore has no time lag.

- Creditable 2 to 5 m per side; the lower limit follows the mapping threshold for field margins (LökPlan GbR, 2024, p. 147).
- No longer farmed as part of the cultivated alley; no soil disturbance after establishment; care within extensive management remains possible.
- Natural regeneration permitted; what counts is a form rich in structure and species. Otherwise 8 points.

6 Calculation and time lag

Core per m² = target value ÷ time lag − initial value
Fringe per m² = 16 − initial value
Compensation value = sum of enhancement × sub-area

Time-lag factor by development time: 1.0 for under 5 years (fringe), 1.2 for 5 to 10 years, 1.5 for 10 to 30, and 2.0 for more than 30 years (MKUEM, 2021, p. 19). The proposal is 1.2 without and 1.5 with standard trees, as in the Practical Guide's worked example of an equipment shed, for a hedge with standard trees (MKUEM, 2021, pp. 105ff.). The whole target value is divided by the factor. Upgrades and downgrades within the discretionary range remain possible, except for features already covered by a value axis.

Worked example

1,000 m of tree strips, 14 m wide including a 2 m fringe on each side: core (between the outer tree rows) 10,000 m², fringe 4,000 m². Mosaic, multi-row structure, species diversity, and harvest interval of 10 years or more: 8 + 1 + 1 + 2 + 3 = 15 points; without standard trees ÷ 1.2 = 12.5.

	Arable land (6)	**species-rich grassland (14)**
Core	6.5 × 10,000 = 65,000	−1.5 × 10,000 = −15,000
Fringe	10 × 4,000 = 40,000	2 × 4,000 = 8,000
Value points	**105,000**	**−7,000**

On species-rich grassland there is no enhancement: the project would itself count as an impact, not as compensation.

7 Commitment, legal safeguarding, and proof of development

- Minimum retention period 20 years, standard trees 50 years, uniform for all tree species; extension possible.
- Safeguarding applies for as long as the impairment lasts (§ 15(4) BNatSchG). For a permanent impact, an extension or a change of site must be planned.
- Safeguarding under § 5 LKompVO: in rem (e.g., easement) on third-party land, otherwise by contract. Under § 19(2) GAPKondV, agroforestry trees are exempt from the prohibition of removal and are therefore not protected against removal under agricultural law.
- The thresholds are binding; if losses take the planting below them, trees are replanted. Above the thresholds, losses of up to 10% are tolerable on large sites, but not automatically on small plantings.
- The stand is confirmed by the proof of development, which the authority as a rule requires after 3 and 5 years.

8 Limitations and open points

- Base values are set by the federal state alone. Until the list is updated, the proposal covers individual cases by analogy and within the discretionary range.
- Derived from official regulations and checked for plausibility in an expert discussion with administrative practice; not empirically validated.
- Only the author's own derivation supports the age threshold for the mosaic, the multi-row structure, the 80% of the bank line, and the 5 m upper limit of the fringe.
- The gap to the old autochthonous hedge remains; the forest criterion “natural cycle” is not credited.
- Time lag 1.5 or 2.0 for standard trees: the guidance on the BKompV (BfN & BMU, 2021) and the Practical Guide differ; to be clarified in the pre-consultation.
- Whether maintenance funding under eco-scheme 3 can be claimed in addition to compensation is open.

Basis: Weitz, J. (2026). Bewertung und kommunale Initiierung von Agroforst über produktionsintegrierte Kompensation in Rheinland-Pfalz [Assessing agroforestry and its municipal initiation through production-integrated compensation in Rhineland-Palatinate] [Master's thesis, Universidade de Lisboa]; developed at the Institute for Applied Material Flow Management (IfaS), Environmental Campus Birkenfeld. The thesis sets out the derivation, the table of stipulations, and two case studies; in addition, there is the rough-estimate calculator “Agroforestry as a compensation measure”: https://julia-nora-weitz.github.io/agroforst-als-kompensation/?lang=en. The proposal replaces neither a compensation balance prepared by an assessor nor consultation on the individual case.

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